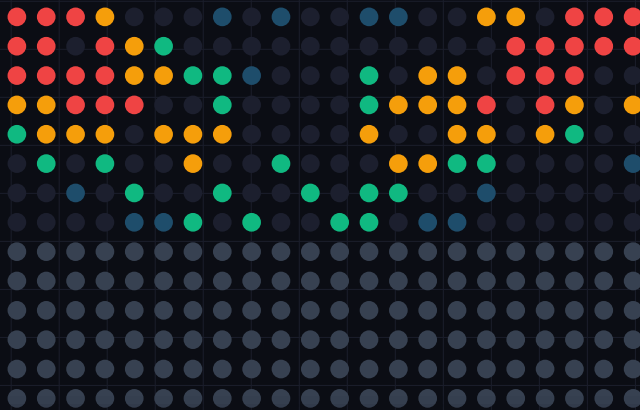


● High ● Medium ● Low ● Empty



CROWD CONTROL

Real-Time Occupancy Intelligence Platform

Project Blueprint · Architecture · Tech Stack · Roadmap

- B2B
- Consumer
- CV + AI
- Real-Time

Platform Overview

What it is, what it does, and who it serves

Live Dashboard

- Real-time occupancy %
- Crowd density heatmaps
- Multi-camera monitoring
- Busy/Not Busy status
- Green-dot person overlay

B2B

Predictive Analytics

- Peak hour prediction
- Wait time estimation
- Crowd surge forecasting
- Historical trend analysis
- Capacity planning alerts

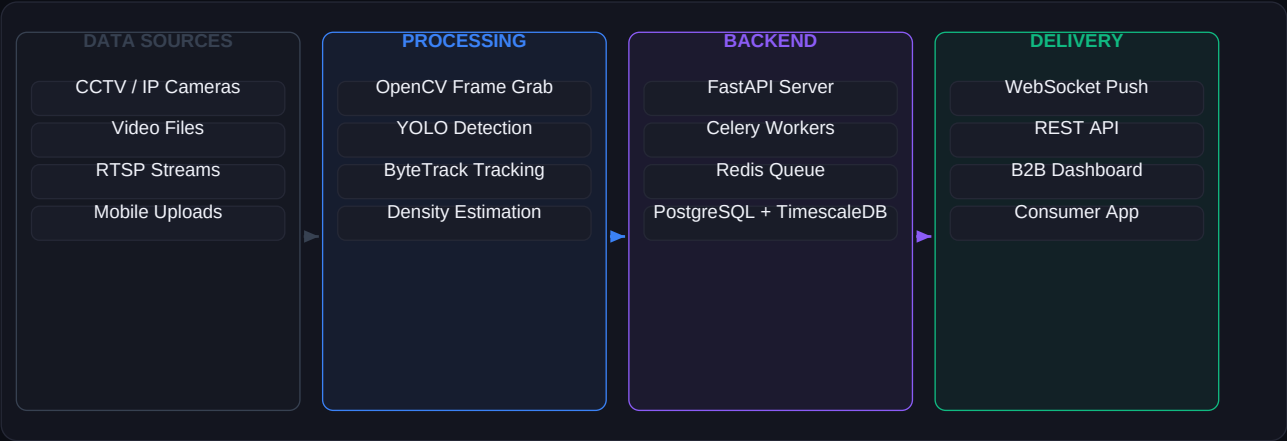
B2B + Consumer

Consumer Discovery

- Check gym crowd live
- Avoid busy restaurants
- Find quiet cafes/malls
- Mobile app interface
- Crowd score per venue

Consumer

SYSTEM ARCHITECTURE



Computer Vision Pipeline

How raw video becomes occupancy intelligence

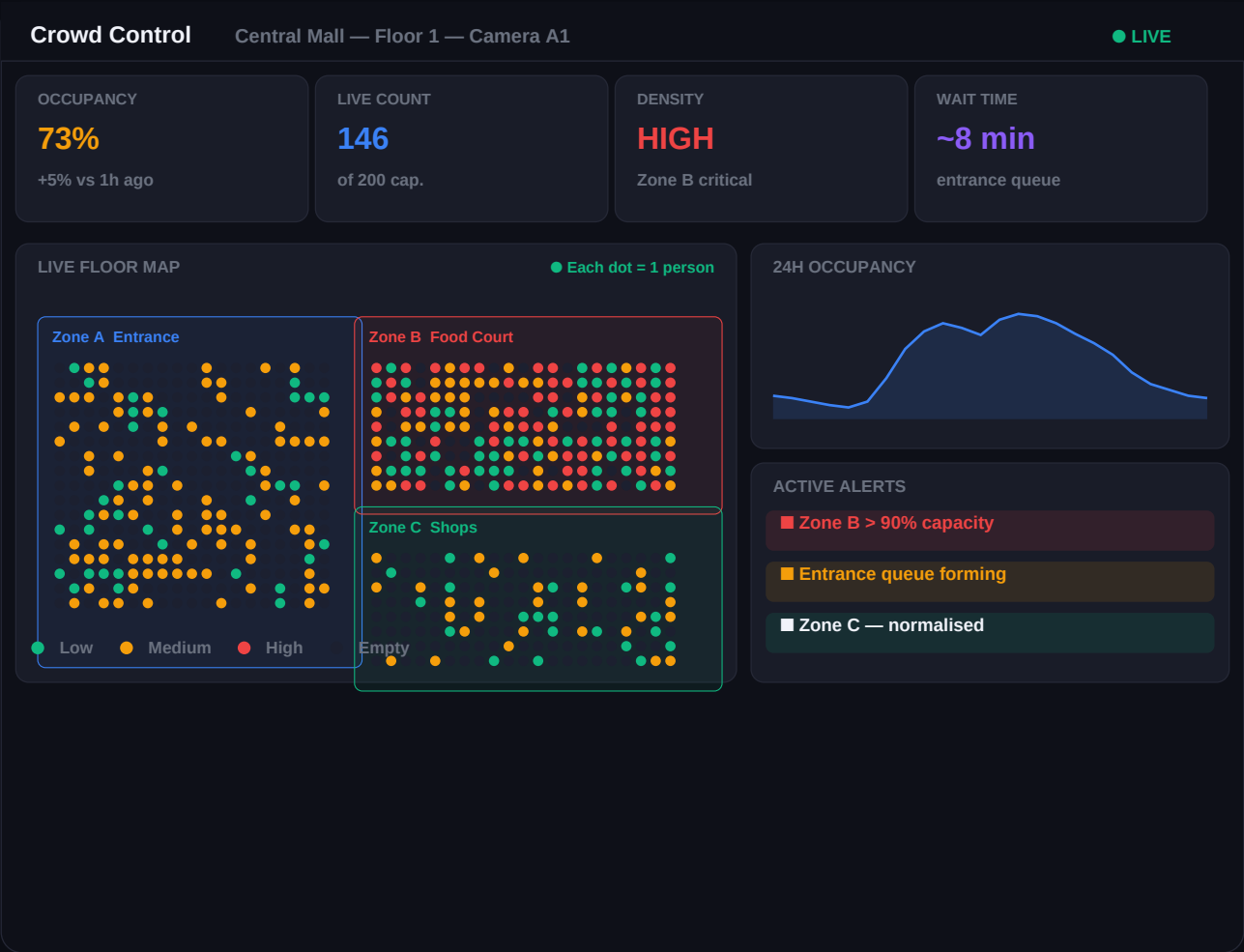


DATABASE SCHEMA

venues		cameras		occupancy_events		predictions	
id	UUID PK	id	UUID PK	id	UUID PK	id	UUID PK
name	VARCHAR	venue_id	UUID FK	camera_id	UUID FK	venue_id	UUID FK
address	TEXT	rtsp_url	TEXT	timestamp	TIMESTAMPZ	predicted_for	TIMESTAMPZ
capacity	INT	zone_name	VARCHAR	count	INT	predicted_count	INT
timezone	VARCHAR	resolution	VARCHAR	occupancy_pct	FLOAT	model_version	VARCHAR
plan_type	ENUM	active	BOOL	density_lvl	ENUM	created_at	TIMESTAMPZ

Live Dashboard Mockup

B2B operator view — real-time occupancy visualization



Tech Stack & API Design

Libraries, services, and endpoint structure

Frontend (React)	Backend (FastAPI)	CV Engine	Infrastructure
React 18 + Vite SPA framework Zustand State management Recharts Analytics charts Leaflet.js Venue floor maps Socket.IO client Live WebSocket feed Tailwind CSS UI styling	FastAPI Async REST + WS Celery + Redis CV task queue SQLAlchemy ORM layer TimescaleDB Time-series data JWT / OAuth2 Auth Alembic DB migrations	OpenCV Frame capture + pre-proc YOLOv8-nano Person detection ByteTrack Multi-object tracking NumPy + SciPy Heatmap generation PyTorch Model inference FFmpeg Stream transcoding	Docker Compose Local dev AWS ECS CV workers CloudFront Dashboard CDN S3 Video/snapshot store Grafana Internal metrics Sentry Error tracking

API ENDPOINTS

GET /api/venues/{id}/occupancy Live occupancy count + status	GET /api/venues/{id}/heatmap Heatmap grid data (JSON)
GET /api/venues/{id}/predictions ML-predicted crowd for next 24h	POST /api/cameras/stream Register new RTSP stream
WS /ws/venues/{id}/live WebSocket: real-time occupancy push	GET /api/venues/{id}/history Historical occupancy time-series
GET /api/consumer/nearby Consumer: crowd scores for nearby venues	GET /api/consumer/venue/{id} Consumer: venue detail + live status

Roadmap & Go-To-Market

Development phases and product positioning

Phase 1 Foundation	Phase 2 CV Engine	Phase 3 B2B Dashboard	Phase 4 Predictive ML	Phase 5 Consumer App
Wks 1–4	Wks 5–8	Wks 9–12	Wks 13–16	Wks 17–20
- FastAPI skeleton + Pos - OpenCV RTSP frame in - YOLOv8 person detecti - Basic occupancy count - Docker Compose dev s	- ByteTrack multi-object - Zone-based density est - Heatmap generation m - TimescaleDB time-serie - Celery async processin	- React live dashboard U - Green-dot floor map wi - WebSocket real-time fe - Multi-camera managem - Alert & threshold syste	- Historical data aggrega - LSTM peak-hour predic - Wait time estimation m - Crowd surge early warn - Forecast accuracy dast	- Mobile-first React Native app - Nearby venue discovery - Crowd score + live status - Favourite venues & alerts - Public API for third parties

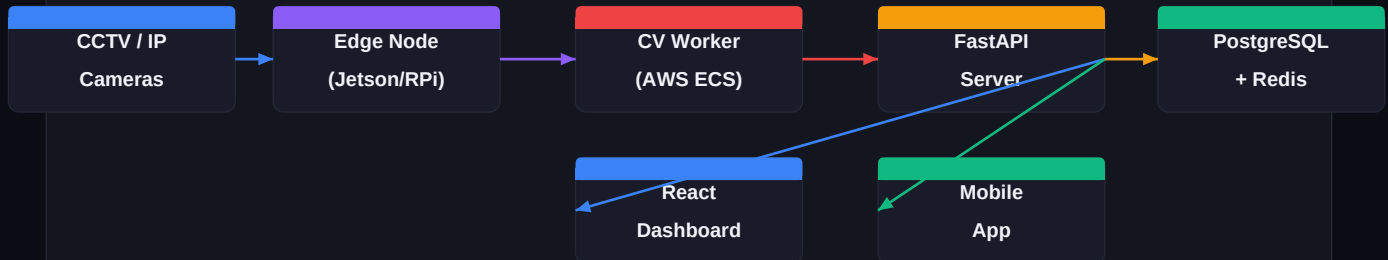
BUSINESS MODEL

B2B — Venue Operators	Consumer — End Users
- SaaS dashboard per venue - CCTV integration + CV processing - Occupancy API for POS / booking systems - White-label option for large chains - Compliance reports (fire code capacity)	- Free app with live crowd scores - Premium: real-time alerts + predictions - Venue discovery by crowd level - Works for gyms, malls, cafes, transit - Monetise via venue partnerships
£99–£499/month per venue	Free + £3.99/month premium

Infrastructure, Privacy & Setup

Deployment, data ethics, and getting started

DEPLOYMENT ARCHITECTURE



PRIVACY & COMPLIANCE

- No facial recognition — count silhouettes only
- Process frames locally on edge; transmit counts, not video
- GDPR: no personal data stored; aggregate counts only
- Signage requirements: venues must display CV monitoring notice
- Video streams never stored — only processed in-memory
- User consent for location data in consumer app

QUICK START

```
bash — setup

# 1. Clone repository
git clone https://github.com/your-org/crowd-control && cd crowd-control

# 2. Start services (Docker)
docker-compose up -d # PostgreSQL + Redis + FastAPI + Celery

# 3. Install Python deps & run migrations
pip install -r backend/requirements.txt && alembic upgrade head

# 4. Add first camera stream
curl -X POST /api/cameras/stream -d '{"rtsp_url": "rtsp://...", "venue_id": "..."}'
```